

Assignment #1

Regression, Gradient Descent, and Regularization

Assigned: 9/1/2026
Due: 9/15/2026, 11:59 PM
Total points: 50

DO NOT USE AI TO WRITE YOUR CODE.

This assignment uses the **Wine Quality** dataset, which will be uploaded to Canvas. Detailed information about this dataset is available at:

<https://archive.ics.uci.edu/dataset/186/wine+quality>

This data set contains data for both red and white wines. Please use the combined data set. Introduce an additional, binary feature *is_red* which will take the value 1 if the wine is red, 0 otherwise. The target variable is **quality**.

Tasks

For this dataset, the following gradient descent algorithms are to be applied to predict wine quality via linear regression:

Method	Variant	Points
Batch gradient descent	without regularization	5
	with L2 regularization (ridge)	5
	with L1 regularization (lasso)	5
Mini-batch gradient descent	without regularization	5
	with L2 regularization (ridge)	5
	with L1 regularization (lasso)	5

Mini-batch means the batch is a stochastically-selected subset of the training data at each iteration (i.e. mini-batch SGD).

Step 0: Inspect and prepare the data

Before starting the ML portion, inspect and analyze the dataset (e.g. summary statistics, distributions, missing values, feature scales). Randomly divide the data into **three** disjoint subsets:

- (a) **Train** (approx. $t\%$): $\mathbf{X}_{\text{train}}$, an $m_{\text{train}} \times (n + 1)$ matrix.
- (b) **Validation** (approx. $v\%$): $\mathbf{X}_{\text{valid}}$, an $m_{\text{valid}} \times (n + 1)$ matrix.

- (c) **Test** (remaining $(100 - t - v)\%$): $\mathbf{X}_{\text{test}}$, an $m_{\text{test}} \times (n + 1)$ matrix.

Here n is the number of attributes (features), so each row has $n + 1$ entries once a constant 1 is prepended for the bias/intercept term w_0 . A typical split is $t \approx 70$, $v \approx 15$ (leaving 15% for test), but you may choose your own reasonable split — just document it.

Save these subsets and reuse them for every task below, so that all six models (and the regularization comparisons) are trained and evaluated on *exactly* the same data.

Notation used throughout:

- $\mathbf{Y}_{\text{train}}$: the ground-truth quality values for the training set.
- $\mathbf{Y}_{\text{valid}}$: the ground-truth quality values for the validation set.
- $\mathbf{Y}_{\text{test}}$: the ground-truth quality values for the test set.
- $\hat{\mathbf{Y}}$: the model's predicted output (e.g. for the training set, $\hat{\mathbf{Y}}_{\text{train}}$).

Step 1: Train and test each model without regularization

Recall the batch gradient descent steps:

1. Let m_{train} be the size of the training set. Set up a weight vector $\mathbf{w}$ of dimension $n + 1$, where n is the number of attributes.
2. Randomly initialize $\mathbf{w} = [w_0, w_1, \dots, w_n]$.
3. Compute the predicted output vector $\hat{\mathbf{Y}}$ of dimension $m_{\text{train}} \times 1$ for the training data: for each data point (row $\mathbf{x}_i$ of $\mathbf{X}_{\text{train}}$), compute $\mathbf{x}_i \cdot \mathbf{w}$.
4. Compute the training MSE as the mean of $(\mathbf{Y}_{\text{train}} - \hat{\mathbf{Y}})^2$ (elementwise square, then average).
5. Compute ∇MSE , the gradient vector of dimension $n + 1$:

$$\nabla \text{MSE}(\mathbf{w}) = \frac{2}{m_{\text{train}}} \mathbf{X}_{\text{train}}^{\top} (\mathbf{X}_{\text{train}} \mathbf{w} - \mathbf{Y}_{\text{train}})$$

6. Update the weights:

$$\mathbf{w} \leftarrow \mathbf{w} - \lambda \nabla \text{MSE}(\mathbf{w}), \quad \text{where } \lambda \text{ is the learning rate.}$$

7. Repeat Steps 3–6 until a termination condition is met (error below some small threshold, a bound on the number of epochs, or a combination of these).

For **mini-batch gradient descent**, carry out the same steps, but with $\mathbf{X}_{\text{train}}, \mathbf{Y}_{\text{train}}$ in Steps 3–6 replaced by a stochastically-selected subset (mini-batch) of the training set at each iteration.

Use this same ∇MSE (boxed above) in each of the models below.

Step 2: L2 regularization (ridge)

Here we seek the weight vector of dimension $n + 1$ that solves:

$$\mathbf{w} = \arg \min_{\mathbf{w}} \left\{ \text{MSE}(\mathbf{w}) + \alpha \|\mathbf{w}\|^2 \right\}.$$

Taking the gradient of the objective:

$$\begin{aligned}\nabla(\text{MSE}(\mathbf{w}) + \alpha \|\mathbf{w}\|^2) &= \nabla \text{MSE}(\mathbf{w}) + \alpha \nabla(\|\mathbf{w}\|^2) \\ &= \nabla \text{MSE}(\mathbf{w}) + 2\alpha \mathbf{w}.\end{aligned}$$

You will need to come up with a way to select the regularization strength α (e.g. using the validation set).

Eliminate the attribute (column of $\mathbf{X}$) whose corresponding weight has the smallest absolute value (i.e. $\min_j |w_j|$, $j \geq 1$, excluding the bias term w_0).

Retrain, on the reduced set of attributes selected by the L2-regularization step above, using **batch** gradient descent on $\mathbf{X}_{\text{train}}$, $\mathbf{X}_{\text{test}}$, etc. as previously saved.

Step 3: L1 regularization (lasso)

Recall $\mathbf{w} \cdot \mathbf{x} = w_0 + w_1x_1 + \dots + w_nx_n$.

For L1 regularization on batch gradient descent, instead of ∇MSE in Step 1.6, use the vector of dimension $n + 1$ with components:

$$\nabla \text{MSE}(\mathbf{w}) + 2\alpha \text{sign}(\mathbf{w}).$$

As with L2, you will need to come up with a way to select α (e.g. using the validation set).

Eliminate the attribute(s) whose corresponding weight(s) shrank closest to zero under L1 regularization.

Retrain, on the reduced set of attributes selected by the L1-regularization step above, using **batch** gradient descent on $\mathbf{X}_{\text{train}}$, $\mathbf{X}_{\text{test}}$, etc. as previously saved.

Deliverables

1. **Figures (10 points).** For *batch* gradient descent, plot the regression line for the attribute corresponding to the *largest* component of $\mathbf{w}$, both before and after each of the L1 and L2 regularization steps.
Note: most likely, the regression lines will *not* look nice — that is expected with a single-attribute view of a multivariate fit.
2. **Written analysis (10 points).** Turn in an analysis (up to one page) comparing the L1 and L2 regularization approaches: e.g. which attributes each one eliminated, how training/validation/test MSE compared across all six models, how you selected α , and any other observations.
3. **Submission.** Turn in your homework in a zipped folder identified by the names (in alphabetical order) of your team members.
4. **Attribution header.** At the top of your program files, list the names of your team members, and either the statement “*all team members have contributed in equal measure to this effort,*” or, if that is not true, state which team members contributed — they will receive full credit, while non-contributing members will have their group grade overridden with a 0.
5. **Comments.** Carefully comment your program.

Hint: If you are coding in MATLAB, MATLAB has a “publish” option that can produce `.doc`, `.pdf`, `.html`, or `.tex` files from your commented script, which can then be further edited.
